

LIBRARY MANAGEMENT SYSTEM

A Java-Based Command-Line Application for Library Automation

Sahil

Registration No. 25BAI10298

VIT Bhopal University | CSE2006 Java Programming

September 16, 2026

Contents

1 Introduction.....	3
2 Features Overview.....	3
2.1 Book Catalog Management.....	3
2.2 Member Management.....	3
2.3 Issue & Return Workflow.....	3
2.4 Fine Calculation.....	3
2.5 Data Persistence.....	3
3 User Guide.....	4
3.1 Adding a Book.....	4
3.2 Registering a Member.....	4
3.3 Issuing a Book.....	4
3.4 Returning a Book.....	4
3.5 Viewing Overdue Books.....	5
3.6 Exiting the Application.....	5
4 Technical Details.....	5
4.1 System Requirements.....	5
4.2 Installation and Execution.....	5
5 Summary Table of Features.....	6
6 Conclusion.....	6

1 Introduction

Library Management System is a Java-based application developed to assist librarians and library members in organizing and managing the day-to-day operations of a small library. Designed with simplicity and reliability in mind, the application offers a clear command-line interface that allows users to catalog books, register members, issue and return items, and automatically calculate late fines. This tool aims to remove the manual overhead of tracking loans on paper, helping libraries maintain accurate, up-to-date records with minimal effort.

2 Features Overview

The core functionalities of the Library Management System revolve around five major operations: cataloging books, registering members, issuing and returning loans, calculating overdue fines, and persisting all records to disk. Each feature is crafted to ensure accuracy and ease of use for day-to-day library operations.

2.1 Book Catalog Management

Users can add new books to the catalog, remove existing ones, and search by ISBN, title, or author. The application stores each book's total and currently available copy counts, so it always knows how many copies are free to lend. This feature promotes accurate, centralized record-keeping for the entire book collection.

2.2 Member Management

To accommodate lending, users can register new members with a unique member ID, name, and email address. Each member's record also tracks any outstanding fine, ensuring that lending decisions are always based on up-to-date account status.

2.3 Issue & Return Workflow

The application manages the complete lifecycle of a loan: issuing a book to a member with a 14-day due date, and recording its return. This workflow ensures every copy in circulation is accounted for at all times.

2.4 Fine Calculation

Returning a book after its due date automatically incurs a fine of \$5.00 per day late, added directly to the member's account. A member with any outstanding fine cannot borrow additional books until the balance is cleared, encouraging timely returns.

2.5 Data Persistence

All catalog, membership, and transaction records are saved to plain CSV files after every action and reloaded automatically the next time the program starts, so no data is ever lost between sessions.

3 User Guide

This section details step-by-step instructions on how to use each major feature of the Library Management System, guiding users through the processes of cataloging, lending, and returning books.

3.1 Adding a Book

1. Select option 1 from the main menu.
2. Enter the ISBN of the book.
3. Enter the title and author.
4. Specify the number of copies being added.

Note: If the ISBN already exists in the catalog, the copy count is increased rather than creating a duplicate entry.

3.2 Registering a Member

1. Select option 5 from the main menu.
2. Enter a unique member ID.
3. Enter the member's name and email address.

Note: This member record will then be available for issuing and returning books.

3.3 Issuing a Book

1. Select option 7 from the main menu.
2. Enter the ISBN of the book to be issued.
3. Enter the member ID borrowing the book.

Note: The application confirms the transaction ID and the due date, set 14 days from the date of issue.

3.4 Returning a Book

1. Select option 8 from the main menu.
2. Enter the transaction ID for the loan being returned.

Note: If the book is returned after its due date, the appropriate fine is automatically added to the member's account.

3.5 Viewing Overdue Books

1. Select option 11 from the main menu.
2. The application will list every loan that is currently overdue.

Note: This list helps librarians follow up on books that have not been returned on time.

3.6 Exiting the Application

Select option 0 to save all data and safely exit the Library Management System.

4 Technical Details

Library Management System is implemented purely in Java without reliance on external libraries, ensuring broad compatibility and straightforward deployment across any system with a Java Development Kit installed.

4.1 System Requirements

- Java Development Kit (JDK), version 8 or later recommended.
- Command line interface (Terminal or Command Prompt) for running the application.

4.2 Installation and Execution

1. Download the file LibraryManagementSystem.java from the project repository.
2. Open a terminal or command prompt and navigate to the directory containing the file.
3. Compile the program by entering the command: `javac LibraryManagementSystem.java`
4. Execute the program by entering the command: `java LibraryManagementSystem`

This procedure launches the Library Management System interface, ready for user interaction.

Data files and CSV schema

FILE	COLUMNS
books.csv	isbn, title, author, totalCopies, availableCopies
members.csv	memberId, name, email, outstandingFine
transactions.csv	transactionId, isbn, memberId, issueDate, dueDate, returnDate

5 Summary Table of Features

FEATURE	DESCRIPTION	USER STEPS
Add Book	Record new books with ISBN, title, author, and copy count.	Menu option 1, input ISBN, title, author, copies.
Register Member	Create a new member record.	Menu option 5, input member ID, name, email.
Issue Book	Lend a book to a member with a 14-day due date.	Menu option 7, input ISBN and member ID.
Return Book	Close out a loan and calculate any late fine.	Menu option 8, input the transaction ID.
Pay Fine	Clear a member's outstanding fine balance.	Menu option 9, input member ID.
List Overdue	Display all loans currently overdue.	Menu option 11, view the overdue list.
Exit	Save all data and quit the application safely.	Menu option 0.

6 Conclusion

The Library Management System application stands as an efficient and accessible tool for small-scale library administration. By combining straightforward cataloging with automated lending and fine calculation, it enables librarians to maintain detailed, accurate records with minimal manual effort. Its dependency solely on the native Java environment ensures ease of setup and portability, making it a practical choice for anyone seeking a no-frills library management solution.